



NSW Education Standards Authority

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Centre Number

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Student Number

2025 HIGHER SCHOOL CERTIFICATE EXAMINATION

Design and Technology

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour and 30 minutes
- Write using black pen
- Write your Centre Number and Student Number at the top of this page

Total marks: 40

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 15 marks (pages 5–8)

- Attempt Questions 11–14
- Allow about 35 minutes for this section

Section III – 15 marks (page 9–12)

- Attempt Question 15
- Allow about 40 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1** What is the most important design factor to consider when designing a flower arrangement for a wedding?

 - A. Aesthetics
 - B. Durability
 - C. Ergonomics
 - D. Obsolescence

- 2** Which of the following should a designer prioritise when designing a product for children?

 - A. Using the most affordable materials available
 - B. Selecting materials with high aesthetic appeal
 - C. Ensuring that materials are safe, durable and non-toxic
 - D. Creating a complex user interface for extended engagement and durability

- 3** Which of the following could a designer use to track the progress of a project over time?

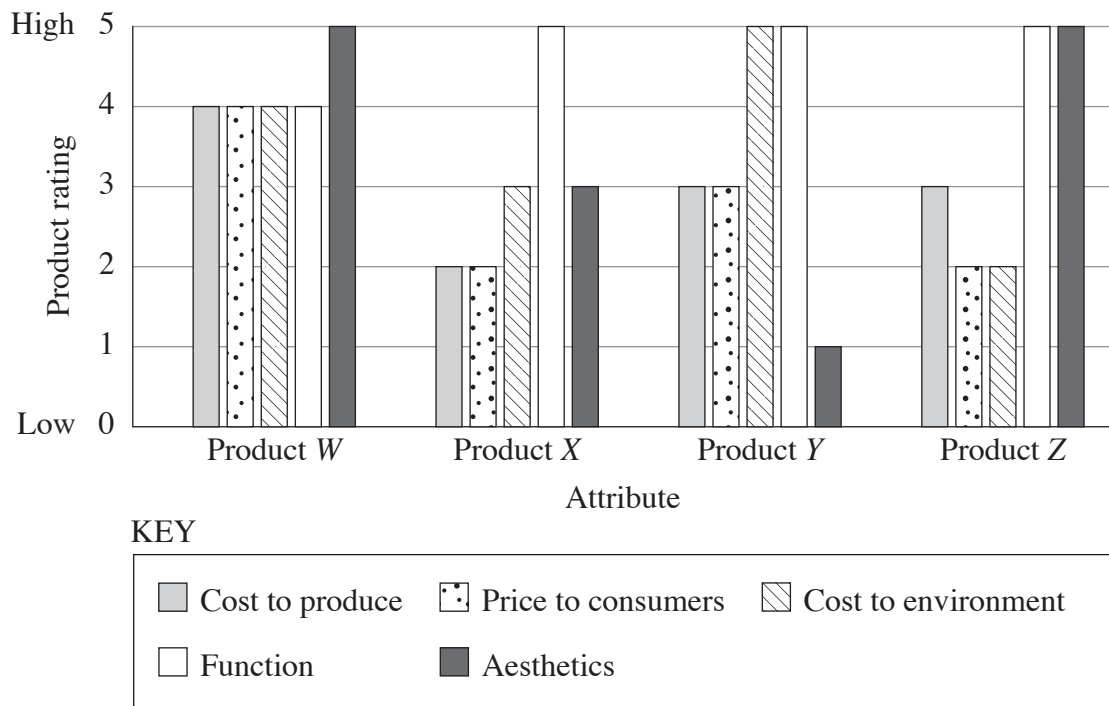
 - A. Gantt chart
 - B. Venn diagram
 - C. SWOT analysis
 - D. Life cycle analysis

- 4** Why is it necessary to experiment and test for durability?

 - A. To check the market demand for the product
 - B. To reduce the environmental impact of the product
 - C. To determine if the product can be sold at a lower price
 - D. To understand how the product functions under different conditions

- 5 Which of the following is the main purpose of analysing data collected when designing a project?
- A. To organise information into tables and charts
 - B. To ensure the data looks visually appealing to investors
 - C. To draw meaningful conclusions that guide decision-making
 - D. To collect as much data as possible about manufacturing trends
- 6 Collaboration with other designers can improve the outcomes of a design project by
- A. reducing the need for communication and feedback.
 - B. ensuring the design project is managed by an expert.
 - C. increasing the manufacturing cost of the design project.
 - D. utilising design experience for different aspects of the project.
- 7 How has packaging design changed as a result of consumer awareness of sustainability?
- A. Increased use of materials that break down over time
 - B. Increased focus on visual appeal through the use of multilayered packaging
 - C. Increased use of non-renewable materials to meet the need of mass production
 - D. Increased use of plastic packaging due to the demand for cheaper options sourced globally
- 8 Which of the following social trends has had the largest impact on the production of new fitness monitoring devices?
- A. Increase in cultural diversity
 - B. Growing importance of preventive healthcare
 - C. Increase in mobile app development standards
 - D. Growing concerns for data privacy and data security

- 9 A company has collected data on four similar products, as shown in the graph.



Which product would be the best choice for consumers to purchase?

- A. W
 - B. X
 - C. Y
 - D. Z
- 10 How has the demand for renewable energy created by emerging technologies had a positive impact on the environment?
- A. By increasing reliance on fossil fuels
 - B. By decreasing greenhouse gas emissions
 - C. By increasing demand for finite resources
 - D. By decreasing demand for renewable resources

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Centre Number

Design and Technology

Sections II and III

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page
- Use this answer booklet for Sections II and III
- Answer BOTH sections
- Follow the instructions inside this booklet as to where you should write your answers

Please turn over

Section II

15 marks

Attempt Questions 11–14

Allow about 35 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 11 (2 marks)

Outline the role of prototyping in the development of a design project.

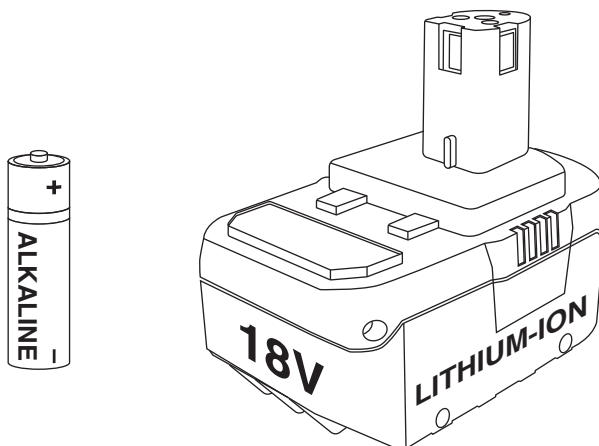
Question 12 (3 marks)

Describe ONE ethical issue that must be considered when managing personal data.

Do NOT write in this area.

Question 13 (4 marks)

TWO different types of battery power options for a household are shown.



Discuss the long-term environmental impact of the battery power options.

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Question 14 (6 marks)

Explain how innovation in design has influenced user experience.

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Do NOT write in this area.

Allow about 40 minutes for this section

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
- communicate ideas and information using relevant examples
- present a logical and cohesive response

Recent advancements in artificial intelligence (AI) have transformed the design industry, influencing how designers create, collaborate, and produce their work. AI tools now assist in generating design concepts, automating repetitive tasks, and providing insights based on data analysis.

Analyse the impact of AI on designers.

[illegible]

– 9 –

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, leaving small margins at the top and bottom. There are no vertical margin lines, text, or other markings on the page.

Do NOT write in this area.

– 10 –

Do NOT write in this area.

This image shows a full page of blank handwriting practice paper. It features approximately 20 horizontal blue lines spaced evenly down the page. Each line is composed of small dots, giving it a dotted or 'dotted-line' appearance. The lines are uniform in color and thickness, providing a guide for letter height and placement. There are no margins, text, or other markings on the page.

– 11 –

[illegible]

– 12 –

Office Use Only – Do NOT write anything, or make any marks below this line.

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